LLM ENGINEERING

RAG: Retrieval-Augmented Generation

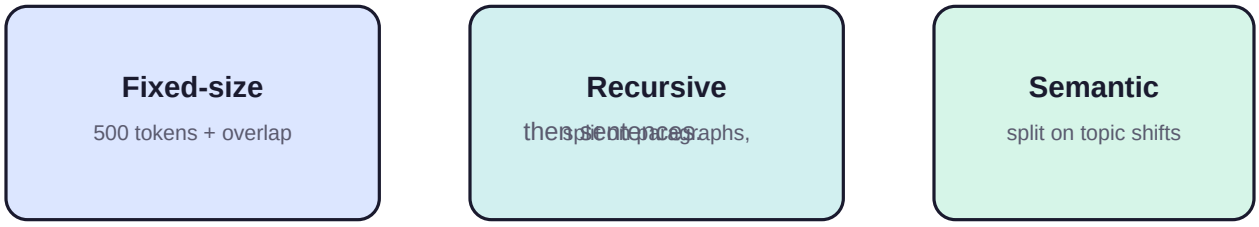
Grounding LLMs in your data:
chunking, retrieval, reranking, cited answers.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

Chunking Strategies

Split documents into retrievable pieces.



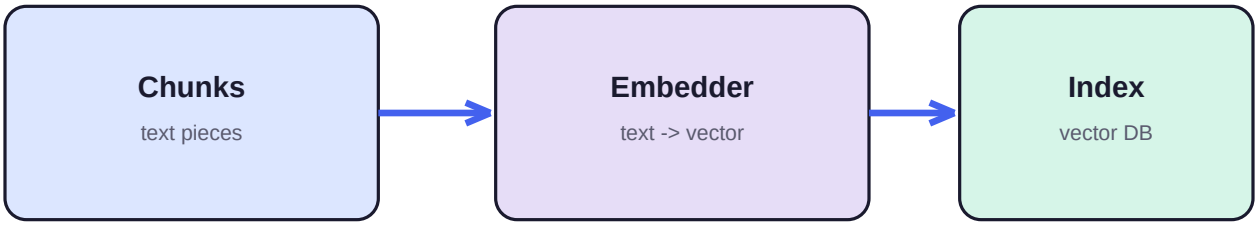
Chunks must be self-contained enough to make sense alone, small enough to rank precisely.

Include metadata (title, page, date) per chunk — it powers filtering and citations later.

STEP 2/6

Embed and Index

Turn chunks into searchable vectors.

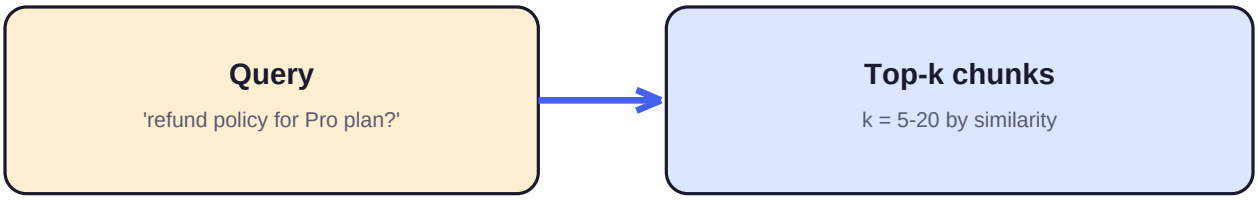


Index once, query many times. Re-embed only when documents change — version your embeddings with your data.

STEP 3/6

Retrieval

Find the chunks that answer the query.



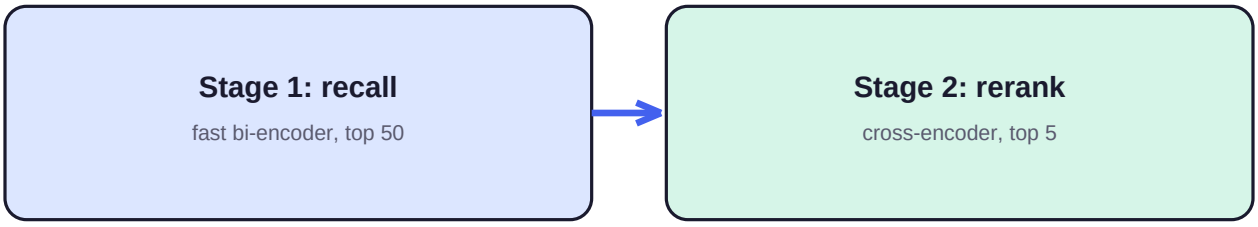
Embed the query with the SAME model, run nearest-neighbor search, apply metadata filters.

Retrieval quality dominates RAG quality. If the right chunk isn't retrieved, the LLM cannot invent it — it will hallucinate instead.

STEP 4/6

Reranking

A second, smarter pass over candidates.

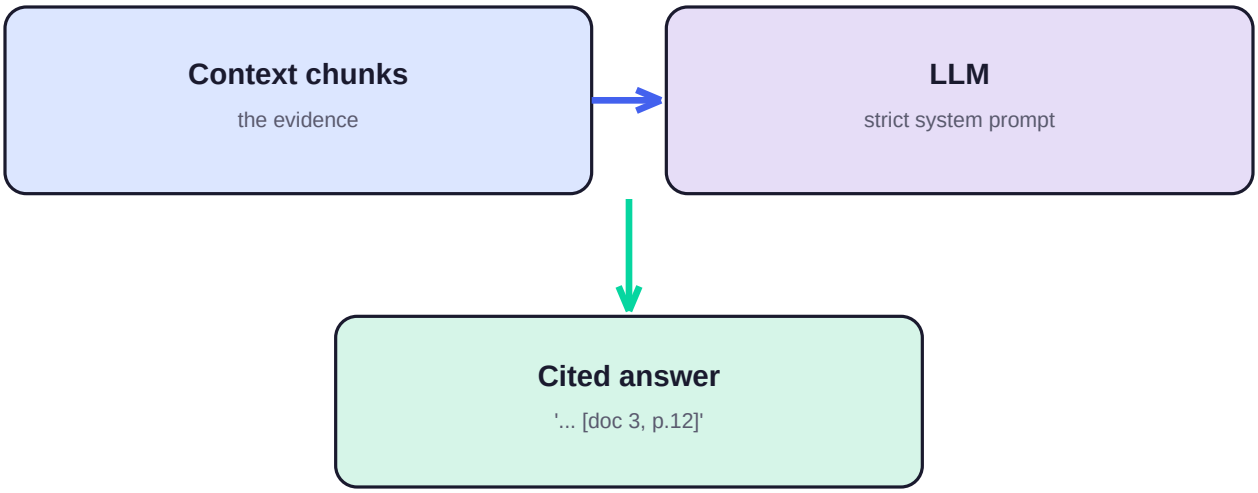


Cross-encoders read query+chunk together — slower but far more accurate. The classic recall-then-precision tradeoff.

STEP 5/6

Generate with Citations

Answer ONLY from retrieved context.



System prompt: answer from context only; cite sources; say 'I don't know' when context is insufficient.

STEP 6/6

Evaluating RAG

Faithfulness first.

- Faithfulness: is every claim supported by a retrieved chunk? (LLM-judge or entailment model).
- Answer relevance: does the answer actually address the question asked?
- Retrieval metrics: recall@k and MRR on a labeled Q/A set — fix retrieval before the prompt.
- Build a golden Q/A set from real user questions; re-run it on every pipeline change.

Most RAG failures are retrieval failures. Log queries with zero good hits — they reveal chunking and coverage gaps.

STEP ✓/✓

Key Takeaways

RAG — Remember

- Chunk for self-containment; attach metadata for filtering and citations.
- Embed once with a consistent model; retrieval quality dominates final quality.
- Rerank with a cross-encoder: recall first, precision second.
- Generate with a strict grounded prompt and require citations; allow 'I don't know'.
- Evaluate faithfulness and retrieval recall on a golden Q/A set, continuously.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev